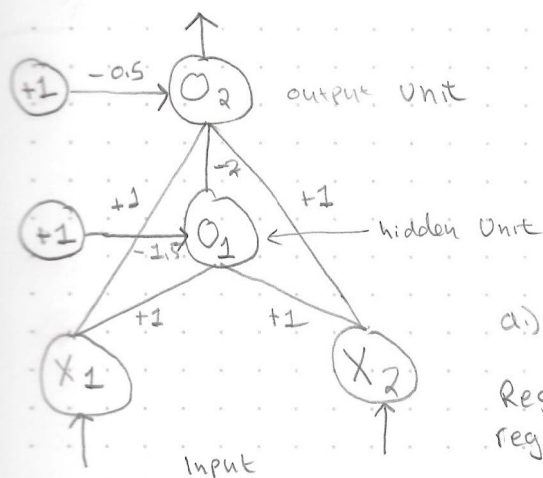


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Homework Assignment #2

Q.) Show that the below network solves XOR by constructing the decision regions for O_1 & O_2 and a truth table for the network

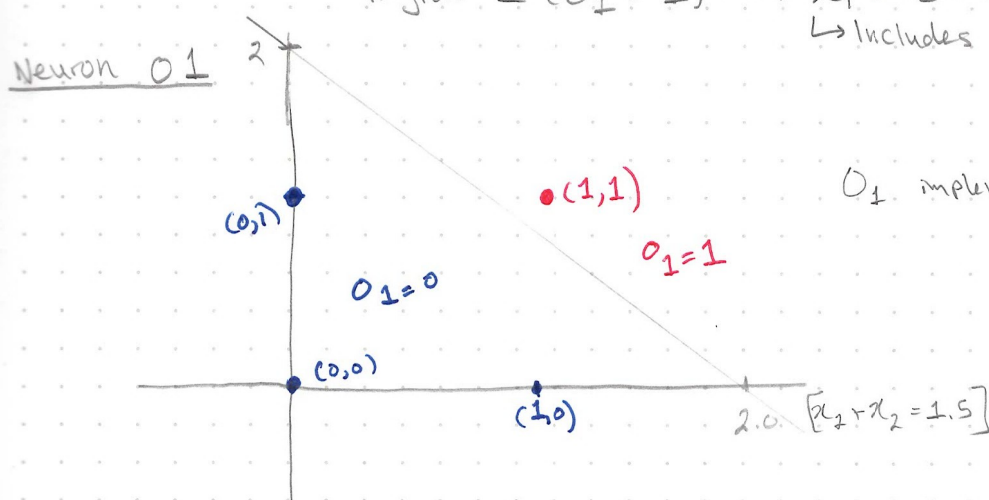


Input	Output
00	0
01	1
10	1
11	0

a) Decision regions for O_1 $\rightarrow x_1 + x_2 - 1.5 = 0$

Region 0 ($O_1 = 0$): Defined by $x_1 + x_2 < 1.5$. This region includes input points (0,0), (0,1), and (1,0)

Region 1 ($O_1 = 1$) $\rightarrow x_1 + x_2 \geq 1.5$
 $\hookrightarrow$ Includes input (1,1)



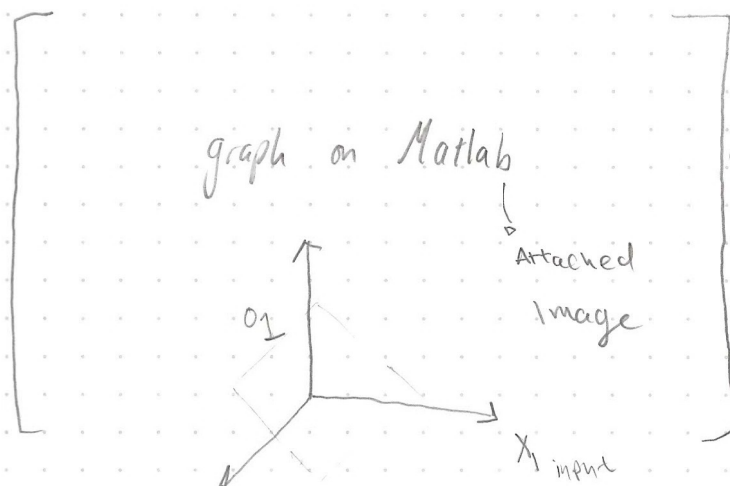
O_1 implements AND Function

b) Decision Regions for O_2 (Output neuron)

$\hookrightarrow$ receives 3 inputs $\rightarrow$ Decision Boundary is a plane in 3D with axes (x_1, x_2, O_1)

$$\rightarrow O_2 = \text{Step}(1 \cdot x_1 + 1 \cdot x_2 - 2O_1 - 0.5)$$

$$0 = x_1 + x_2 - 2O_1 - 0.5 \quad \leftarrow \text{Equation for decision Boundary plane}$$



Points above plane $\rightarrow O_2 = 1$

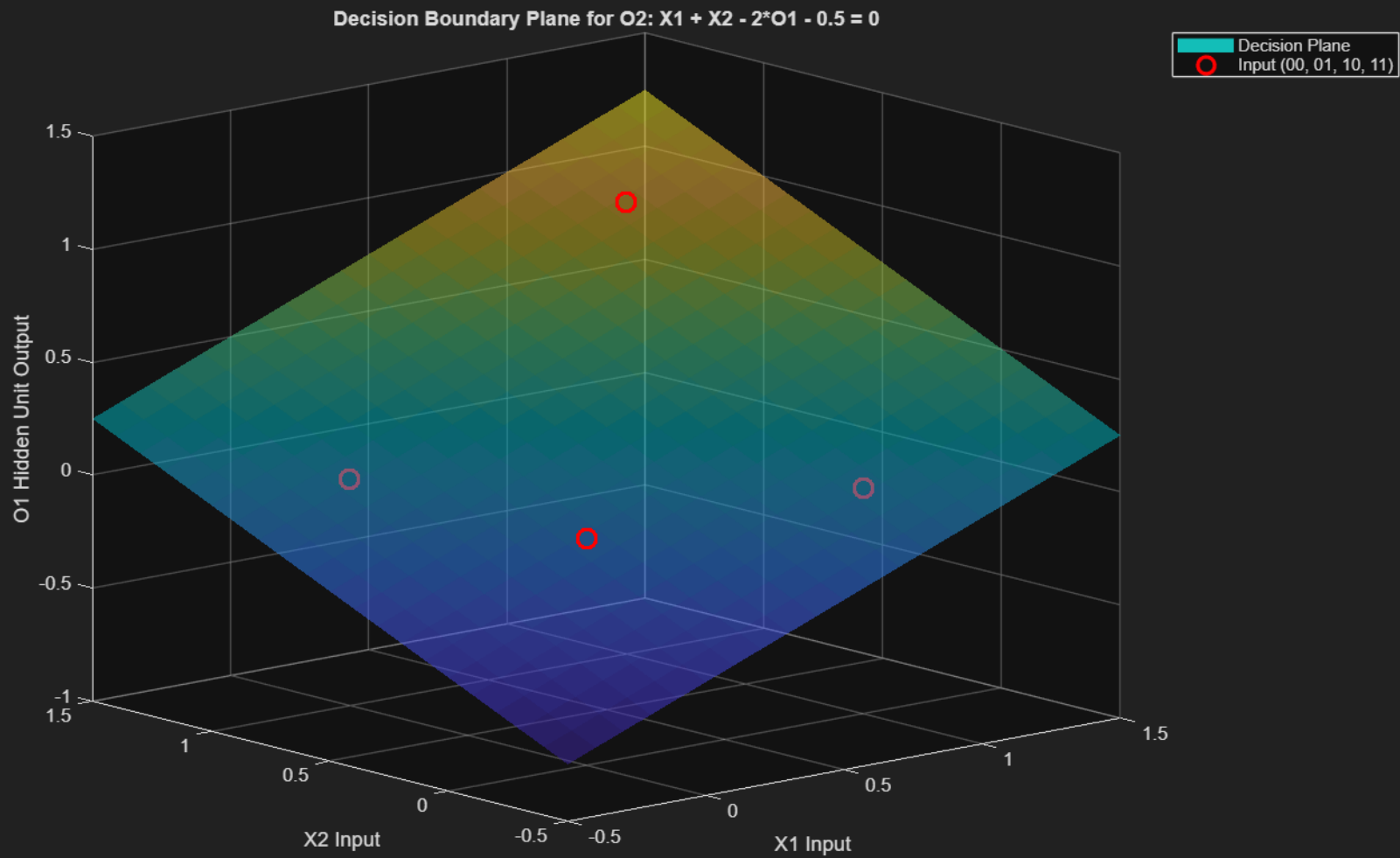
Points below plane $\rightarrow O_2 = 0$

For input (0,0,0) $\rightarrow 0+0-0-0.5 = -0.5$

For input (0,1,0) & (1,0,0) $\rightarrow$ sum is 0.5

For input (1,1,1) $\rightarrow 1+1-2(1)-0.5 = -0.5$

```
plot2.m
1  % Define the grid for X1 and X2
2  [X1, X2] = meshgrid(-0.5:0.1:1.5, -0.5:0.1:1.5);
3
4  % Calculate the decision boundary plane for O1 (Z-axis)
5  % Equation:  $X1 + X2 - 2*O1 - 0.5 = 0 \Rightarrow 2*O1 = X1 + X2 - 0.5 \Rightarrow O1 = (X1 + X2 - 0.5) / 2$ 
6  O1_plane = (X1 + X2 - 0.5) / 2;
7
8  % Plot the plane
9  figure;
10 surf(X1, X2, O1_plane, 'FaceAlpha', 0.5, 'EdgeColor', 'none');
11 hold on;
12 xlabel('X1 Input');
13 ylabel('X2 Input');
14 zlabel('O1 Hidden Unit Output');
15 title('Decision Boundary Plane for O2:  $X1 + X2 - 2*O1 - 0.5 = 0$ ');
16 grid on;
17
18 % Plot the specific input points in (X1, X2, O1) space
19 % Points are formatted as [x1, x2, o1_val]
20 points = [0 0 0; 0 1 0; 1 0 0; 1 1 1];
21 plot3(points(:,1), points(:,2), points(:,3), 'ro', 'MarkerSize', 10, 'LineWidth', 2);
22 legend('Decision Plane', 'Input (00, 01, 10, 11)');
```



c.) Truth table for the network

x_1	x_2	Net Input to O_1 ($x_1 + x_2 - 1.5$)	O_1 output	Net Input to O_2 ($x_1 + x_2 - 2.0 - 0.5$)	O_2 output
0	0	-1.5	0	-0.5	0
0	1	-0.5	0	+0.5	1
1	0	-0.5	0	+0.5	1
1	1	+0.5	1	-0.5	0

Network output

